



DAFFODIL INTERNATIONAL UNIVERSITY
Department of Computing and Information System (CIS)

Algorithm

Lab Report-06

Submitted By:

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Program: B.Sc. in CIS

Submitted To:

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Class_Code

```
#include <stdio.h>
int main(){
    int a[100], n, i, j,min, temp;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    printf("Enter elements: ");
    for(i = 0; i < n; i++){
        scanf("%d", &a[i]);
    }
    for(i = 0; i < n - 1; i++){
        if(a[i] % 2 == 0){
            min = i;
            for(j = i + 1; j < n; j++){
                if(a[j] % 2 == 0 && a[j] < a[min]){
                    min = j;
                }
            }
            temp = a[i];
            a[i] = a[min];
            a[min] = temp;
        }
    }
    printf("Array after sorting even numbers only:\n");
    for(i = 0; i < n; i++){
        printf("%d ", a[i]);
    }
    return 0;
}
```

Output:

```
> cd "i:\Progra
f ($?) { gcc classCode.c -o classCode } ; if ($?) { .\classCode }
Enter the number of elements: 5
Enter elements: 44
23
66
87
56
Array after sorting even numbers only:
44 23 56 87 66
PS I:\Programming\All Coding\Basic C ,C++\3rd semester Algorithm Lab\Lab report 6> █
```

Task 1

```
#include <stdio.h>
int main(){
    float a[100], temp;
    int n, i, j, min;
    printf("Enter the number of students: ");
    scanf("%d", &n);
    printf("Enter CGPAs: ");
    for(i = 0; i < n; i++){
        scanf("%f", &a[i]);
    }
    for(i = 0; i < n - 1; i++){
        min = i;
        for(j = i + 1; j < n; j++){
            if(a[j] > a[min]){
                min = j;
            }
        }
        temp = a[i];
        a[i] = a[min];
        a[min] = temp;
    }
    printf("CGPAs in descending order:\n");
    for(i = 0; i < n; i++){
        printf("%.2f ", a[i]);
    }
    return 0;
}
```

Output:

```
> cd ..1
f ($?) { gcc task1.c -o task1 } ; if ($?) { .\task1 }
Enter the number of students: 4
Enter CGPAs: 3.76
3.88
3.50
3.92
CGPAs in descending order:
3.92 3.88 3.76 3.50
PS I:\Programming\All Coding\Basic C ,C++\3rd semester Algorithm Lab\Lab report 6> █
```

Task 2

```
#include <stdio.h>
int main(){
    int a[100], n, i, j, min, temp;
    printf("Enter the number of requests: ");
    scanf("%d", &n);
    printf("Enter priority scores: ");
    for(i = 0; i < n; i++){
        scanf("%d", &a[i]);
    }
    for(i = 0; i < n - 1; i++){
        min = i;
        for(j = i + 1; j < n; j++){
            if(a[j] < a[min]){
                min = j;
            }
        }
        temp = a[i];
        a[i] = a[min];
        a[min] = temp;
    }
    printf("Priority scores in ascending order:\n");
    for(i = 0; i < n; i++){
        printf("%d ", a[i]);
    }
    return 0;
}
```

Output:

```
> cd 1
f ($?) { gcc task2.c -o task2 } ; if ($?) { .\task2 }
Enter the number of requests: 4
Enter priority scores: 12
33
22
44
Priority scores in ascending order:
12 22 33 44
PS I:\Programming\All Coding\Basic C ,C++\3rd semester Algorithm Lab\Lab report 6> █
```

End